SEAKEEPER SERVICE TOOL GUIDE - 118

PRODUCT ALL SEAKEEPER 1 THROUGH 40

Contents

PURPOSE

This procedure provides instructions for using the Seakeeper Service Tool Application.

OVERVIEW

Seakeeper Service Engineering manages the distribution of the application's installer and activation keys. The application must be installed and activated using a licensing key specific to each user and device. Installation onto multiple devices requires separate license keys for each. The selection of some buttons and their functionality may be limited based on granted license access level.

The Seakeeper Service Tool Application can perform the following functions:

- Program all Seakeeper electronic devices (GCM, Motor Drive) to the most up to date software version available – ONLY when connected to a Seakeeper CAN network with a GCM running 10.1XX or greater software.
- Display Seakeeper system information the same as seen from a display/MFD's information page.
- Log Seakeeper CAN bus traffic that can be provided to Seakeeper Service in diagnosing issues.
- Identify and provide the latest 5" display or ConnectBox software but CANNOT program these devices. The software must be moved to a USB flash drive and inserted into the user interfaces.
- Allow viewing of alarm history after deleting from the alarm history screen of user interface.
- Allow viewing of hidden **Performance Info** tab in Settings screen [i.e., drive input power, last spool time, last steady-state power, coast power, and reset integral terms (proportional valve setting)].

NOTE:
ConnectBox equipped Seakeepers with the GCM software

TOOLS/SUPPLIES REQUIRED

- Service Tool Hardware Kit (P/N 20557)

REFERENCE

- SWI-137 – ConnectBox Data Log Retrieval

1.0 HARDWARE SETUP

1. **IF** Seakeeper equipped with 5” Touch Display only (Figure 1),
   THEN DISCONNECT wire harness communications cable connector from display cable.

2. **IF** Seakeeper equipped with ConnectBox (Figure 2),
   THEN DISCONNECT wire harness from ConnectBox T-adapter.

3. **INSTALL** T-adapter with Seakeeper CAN to USB device onto wire harness DISPLAY CANbus connector and display cable or ConnectBox T-adapter as shown below.

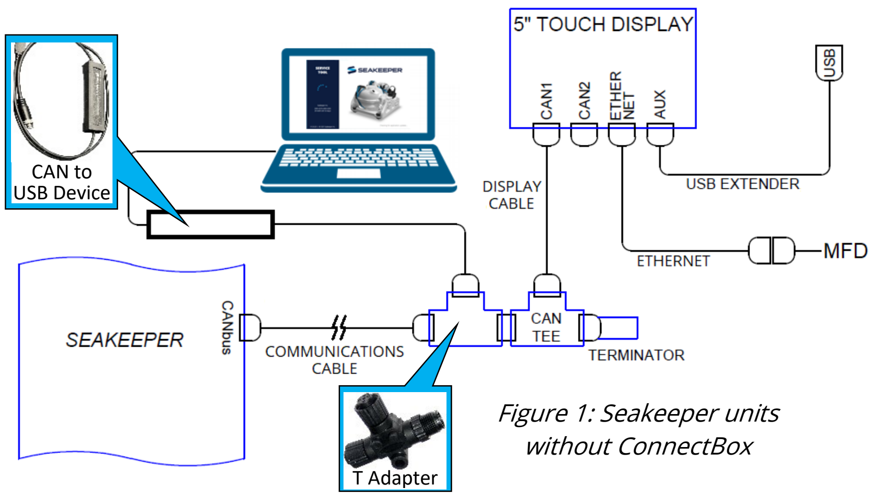


Figure 1: Seakeeper units without ConnectBox

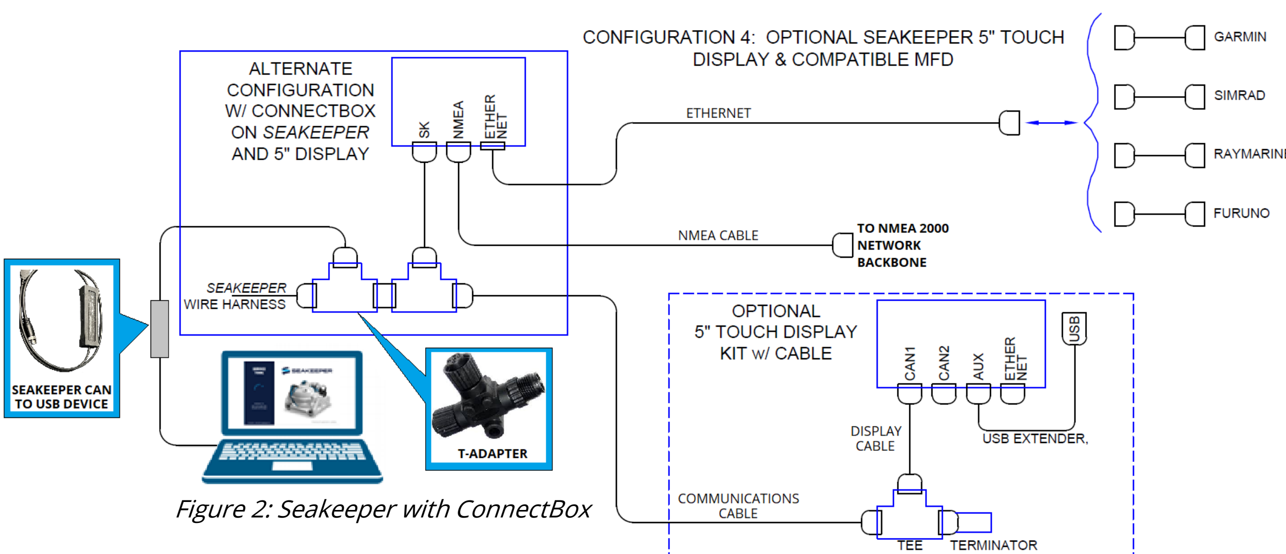


Figure 2: Seakeeper with ConnectBox

2.0 OPENING APPLICATION

1. **PLUG** Seakeeper USB to CAN adapter into laptop. [Adapter does not need to be plugged into Seakeeper for service tool application to launch.]

2. **OPEN** Seakeeper Service Tool App by one of following steps:

- **DOUBLE-CLICK** on desktop icon (Figure 3).
- From Windows START (Figure 4):
    - **SELECT** START button.
    - **TYPE** "Seakeeper Service Tool" in search field.

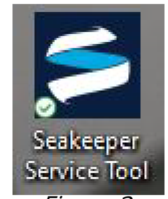


Figure 3: Service tool icon

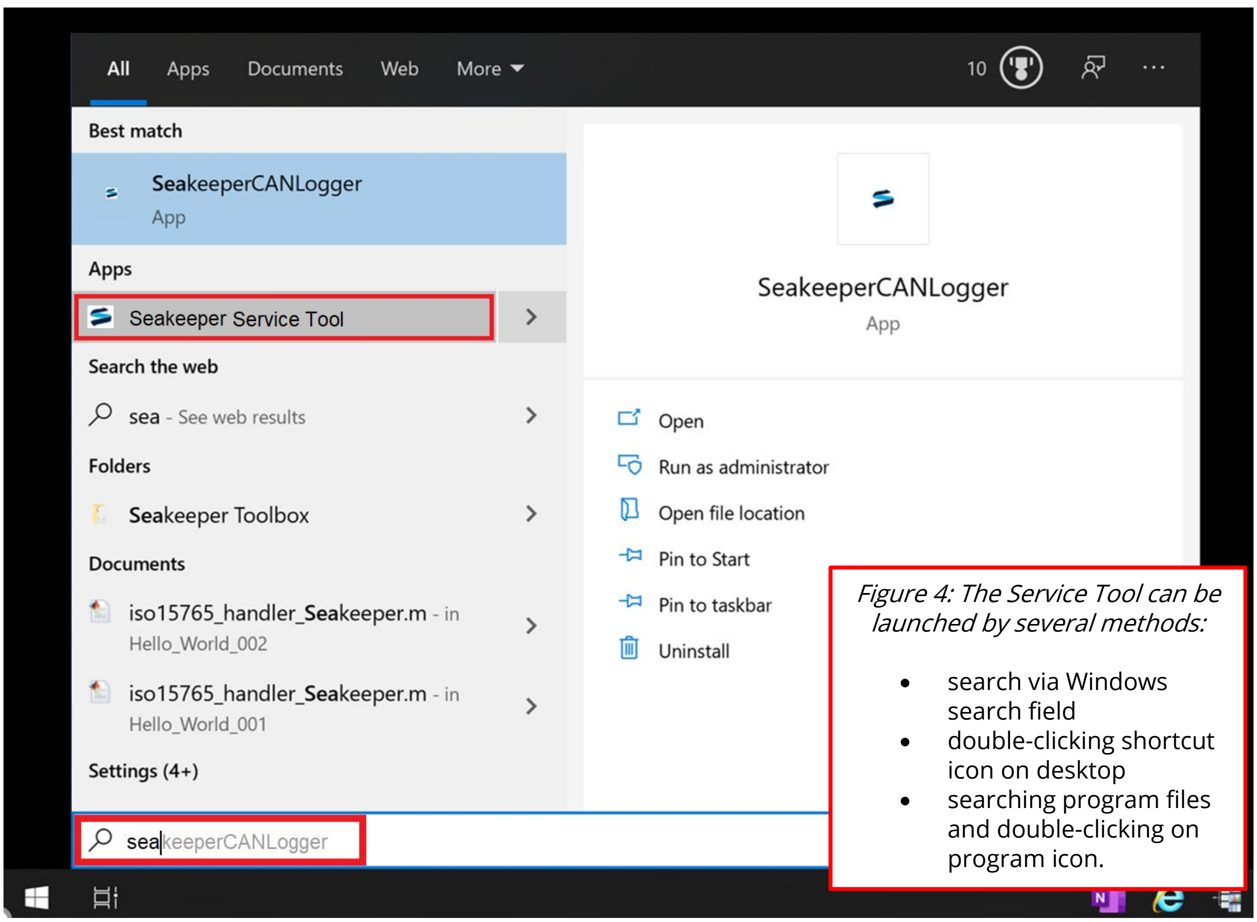


Figure 4: The Service Tool can be launched by several methods:

- search via Windows search field
- double-clicking shortcut icon on desktop
- searching program files and double-clicking on program icon.

NOTE:

Application will attempt to connect via Internet to update possibly causing data charges.

CAUTION:

APPLICATION WILL LOCKOUT if unable to connect to Internet within six days of last usage.

3. **ALLOW** application to synchronize with online server to obtain latest software and permissions. [Figure 5 splash screen appears as service tool attempts to obtain software updates and resets lockout timer.]

4. **<u>IF</u>** Internet unavailable upon start, **<u>THEN</u> ENSURE** sufficient time available before lockout.

Figure 5: Service Tool splash screen

a. **<u>IF</u>** opening app without Internet connectivity and time before lockout remaining, **<u>THEN</u> DISREGARD** pop-ups that may appear by selecting "OK."

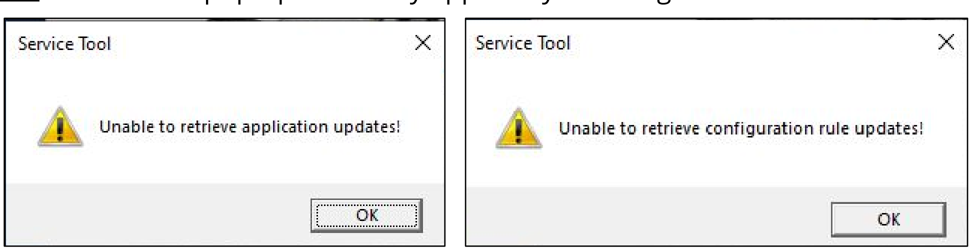


Figure 6: Notifications when no internet detected

5. After splash screen, **ENSURE** Seakeeper CAN to USB Device appears and 250 kHz baud rate selected (figure 7).

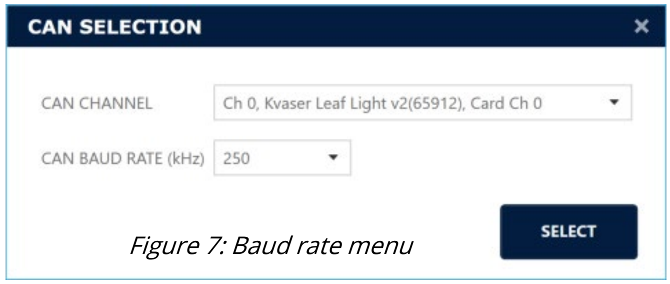


Figure 7: Baud rate menu

6. **PRESS** select button.

   a. **<u>IF</u>** Seakeeper USB to CAN adapter is not plugged in or recognized by the PC [as seen by *NO CHANNELS FOUND* in CAN CHANNEL dropdown], **<u>THEN</u> CORRECT** issue **<u>AND</u> RELAUNCH** service tool.

NOTE:
The Performance Info tab is only available when the Service Tool app is connected.

7. **<u>IF</u>** desired to view **Performance Info** on MFD / 5" Display, **<u>THEN</u>:**

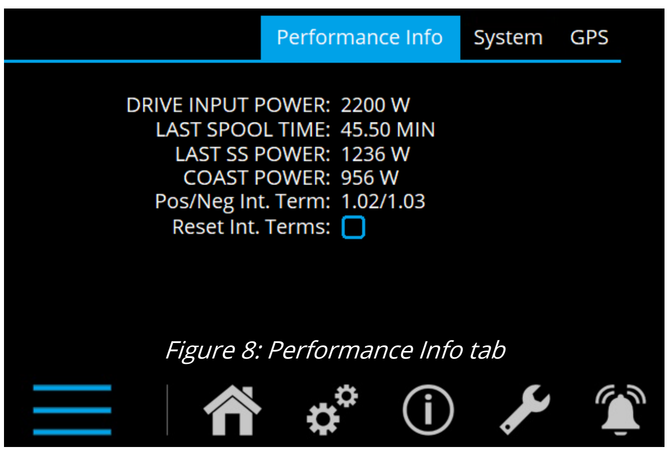


Figure 8: Performance Info tab

   a. **NAVIGATE** to Settings screen (gears icon) of user interface.

   b. **PERFORM** long press of gears icon to access hidden screen.

   c. **SELECT** Performance Info tab to view:

      - Drive Input Power [W] = Drive input Voltage X Drive Input Current
      - Last Spool Time – calculated as time for last acceleration from 100 RPM to (target speed– 100) RPM
      - Coast Power – Mechanical power based on flywheel deceleration when power button stops motor (AC power still applied to drive).
      - Reset Int. Terms – User input to force prop valve integral terms to 1.00 (for diagnostics)

3.0 UPDATING SEAKEEPER SOFTWARE

> **NOTE:**
>
> - If GCM software NOT 10.1XX or later, application will give following error:
>   *Error! This Seakeeper is not capable of field software updates*
> - IMU software earlier than **2.60** cannot be updated by Service Tool app. IMUs with earlier revisions should be replaced.

1. **ENSURE** following:

   a. Seakeeper CAN to USB Device connected to a powered Seakeeper GCM through its Seakeeper CAN.

   b. Seakeeper GCM software version 10.1XX or later.

2. **SELECT** UPDATE menu button (Figure 9).

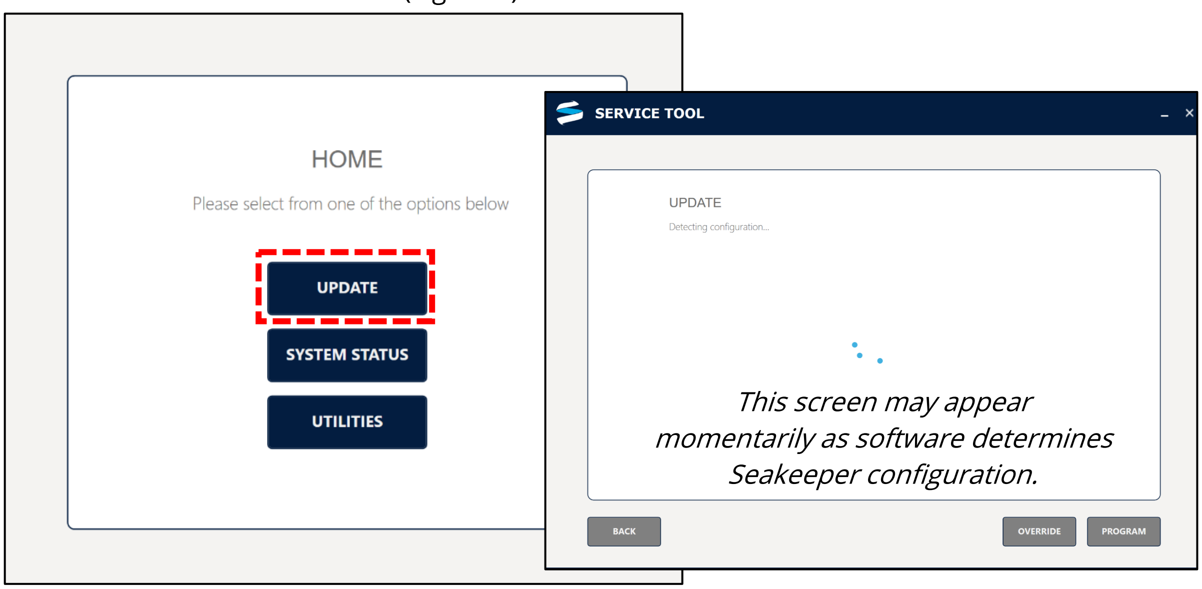
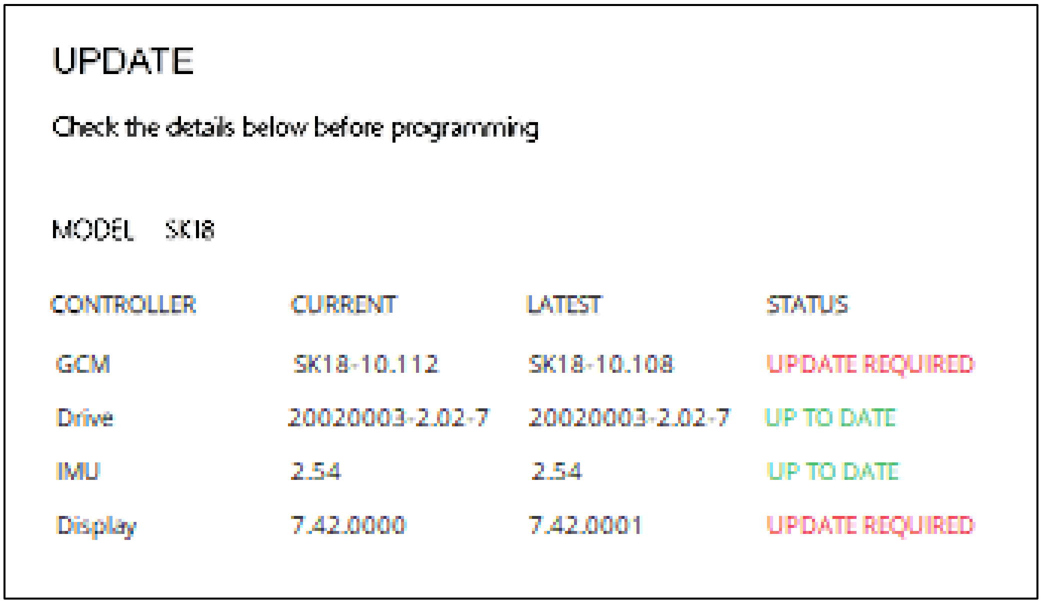


Figure 9: Home screen with Update

3. **REVIEW** software update options (See Figure 11 example).



UPDATE

Check the details below before programming

MODEL SK18

CONTROLLER	CURRENT	LATEST	STATUS
GCM	SK18-10.112	SK18-10.108	UPDATE REQUIRED
Drive	20020003-2.02-7	20020003-2.02-7	UP TO DATE
IMU	2.54	2.54	UP TO DATE
Display	7.42.0000	7.42.0001	UPDATE REQUIRED

Figure 11: Software update options example

4. **<u>IF</u>** expected device is not found (Figure 12),
   <u>THEN</u>:

   a. **CYCLE** DC power to Seakeeper.

> **NOTE**
>
> The tool will detect and allow programming of the GCM and drive even if it does not detect an IMU, display, or ConnectBox.

   b. **CONFIRM** wire harness connectors, T-Adapters, and terminators are securely fastened on associated component(s).

   c. **SELECT** REDETECT button.

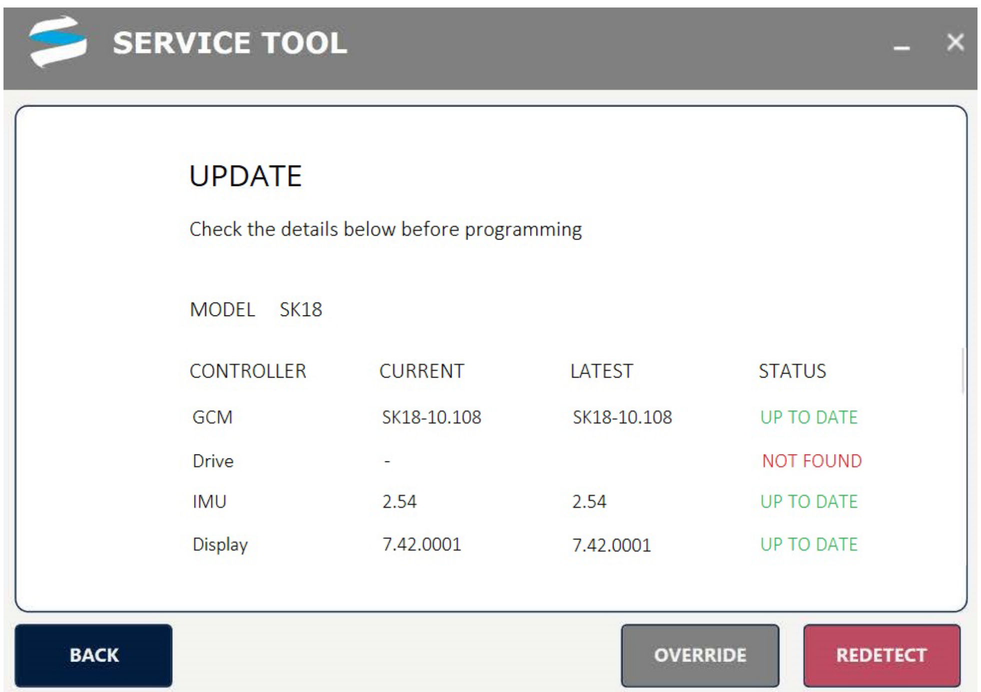


Figure 12: Update screen with Redetect button

   d. **<u>IF</u>** GCM or Drive is still not detected,
      <u>THEN</u>:

      i. **DISCONNECT** IMU, ConnectBox, and 5” display.

      ii. **SELECT** REDETECT button.

NOTE:

Once all devices detected, if any software update(s) available, PROGRAM button can be selected to proceed to below update screen. Service tool will begin programming each device one at a time automatically.

5. **SELECT** PROGRAM button to navigate to UPDATE page (Figure 13).

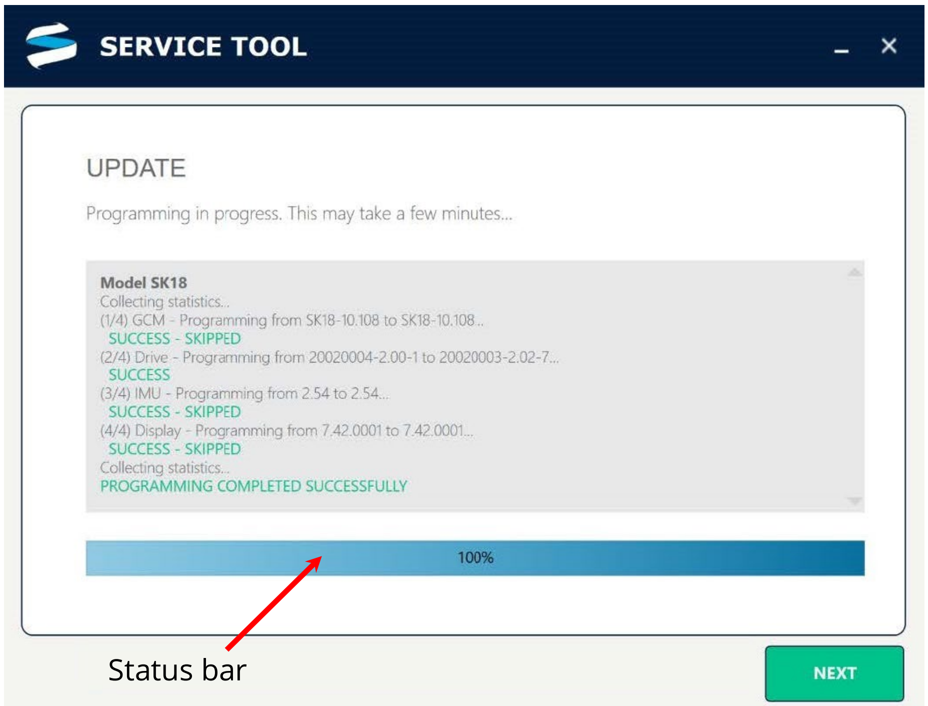


Figure 13:

Devices that showed UPDATE REQUIRED in STATUS column will now be programmed. Devices that showed UP TO DATE will be *skipped*.

6. **<u>WHEN</u>** software update completed (status bar at 100%),
<u>THEN</u> CYCLE DC control power to Seakeeper for update to register.

7. **<u>IF</u>** programming of specific device fails,
<u>THEN</u> CONTACT Seakeeper Product Support Team at support@seakeeper.com.

4.0 DISPLAY / CONNECTBOX PROGRAMMING

1. When window in Figure 14 appears, **INSERT** blank FAT-32 USB flash drive.

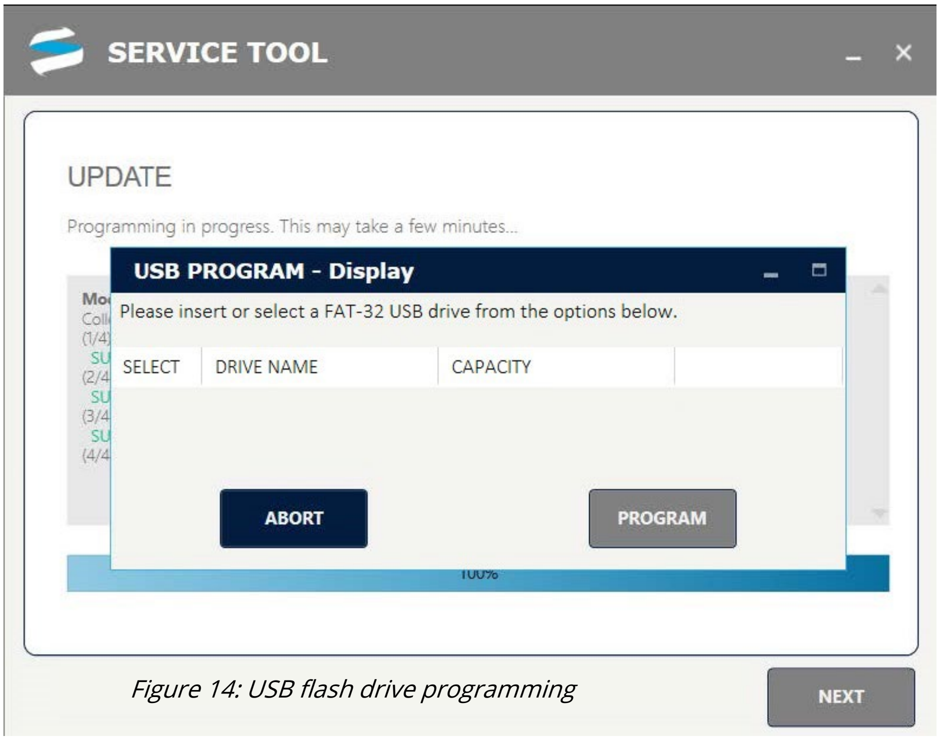


Figure 14: USB flash drive programming

2. Once flash drive appears (Figure 15), **SELECT** flash drive.

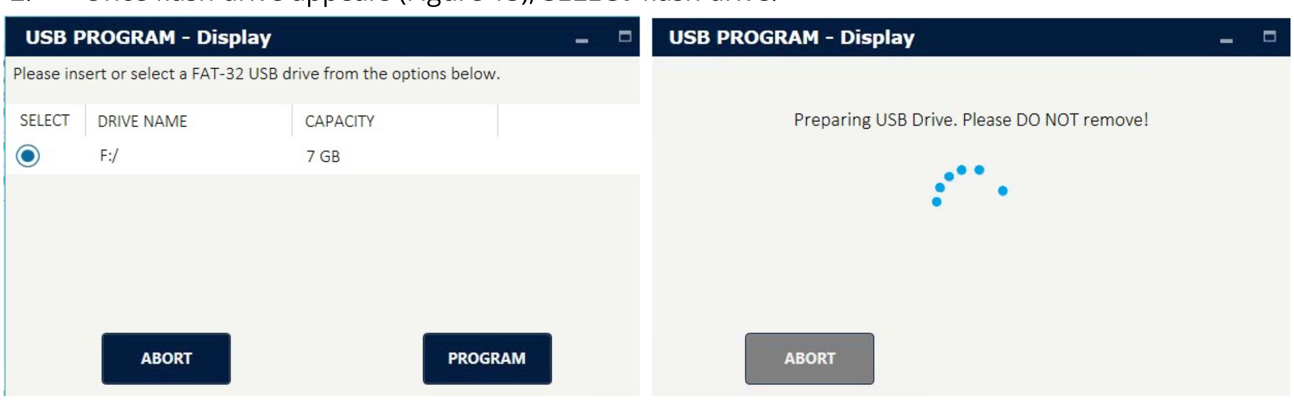


Figure 15: Selecting flash drive location and starting transfer

3. **SELECT** PROGRAM button to program flash drive for display/ConnectBox update.

4. Once latest software moved to USB flash drive, **REMOVE** flash drive when prompted.

5. <u>**IF**</u> updating 5” display,
   <u>**THEN**</u> **CONNECT** USB flash drive directly into back of display to begin update.

Figure 16: 5” display USB port location and update screen

   a. Once 5” display updated and prompted to do so, **REMOVE** USB flash drive.

Figure 17: Removal of flash drive when prompted

6. <u>**IF**</u> updating a ConnectBox,
   <u>**THEN**</u>**:**

   a. **REMOVE** ConnectBox micro-SD port cover using Torx-8 screwdriver.

   b. **FULLY INSERT** micro-SD cable end into ConnectBox micro-SD port.

   c. **PLUG** USB flash drive into OTG cable.

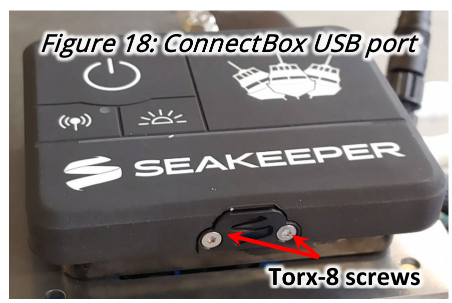


Figure 18: ConnectBox USB port

Figure 19: OTG cable

   d. Without cycling power or removing cable, **WAIT** 5 to 10 minutes as software installs.

7. **VERIFY** Seakeeper Service Tool shows new software after several minutes.

Figure 20: Connecting OTG cable

5.0 SYSTEM STATUS

1. From HOME screen, **SELECT** SYSTEM STATUS button.

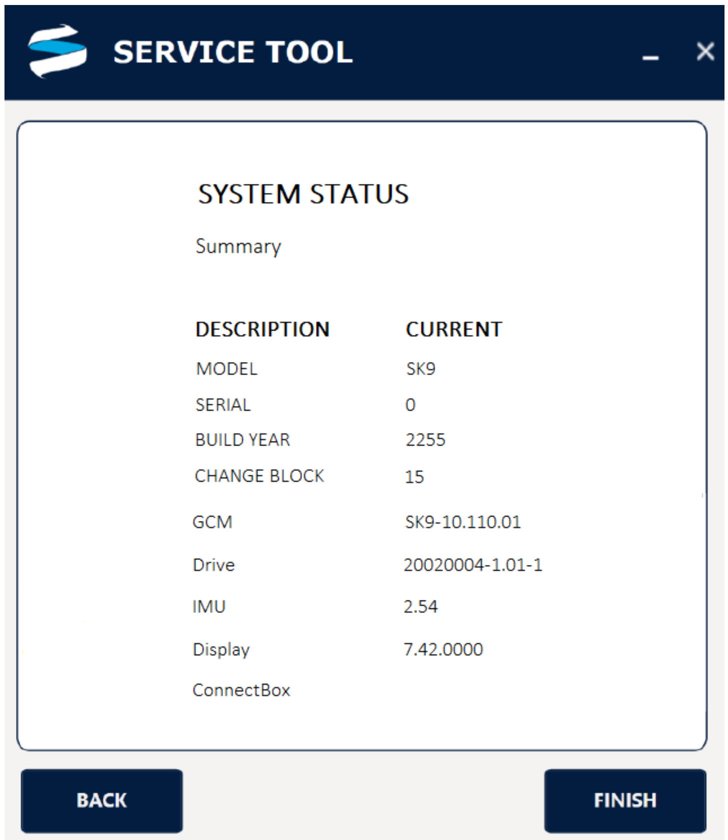


Figure 21: System Status window

2. When review of data complete, **SELECT** BACK or FINISH button to return to HOME screen.

6.0 UTILITIES

Selecting the UTILITIES button from the HOME page provides the screen below.

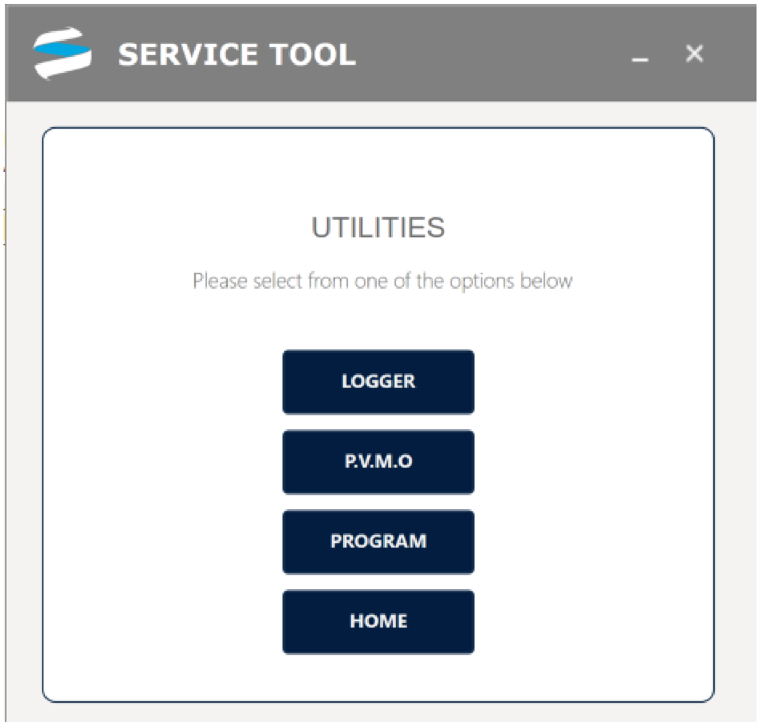


Figure 22: Utilities window menu

6.1 LOGGER

The logger can be used to capture data during all sea trials as a precaution but should only be submitted to Seakeeper Service for additional troubleshooting assistance for the following reasons:

- Alarm codes 37, 38, 39, and 56
- Undesirable clunking, jarring, or shuddering during operation where no alarms are present
- Alarm 163, where unit spools past 150 rpm, sphere vacuum level is good, and there are no indications of bearing failure due to elevated noise or vibration levels
- Underwhelming roll reduction performance not already expected given sizing of Seakeeper to vessel
- Unusual Seakeeper sphere movement, such as moving very little in larger than 2 ft seas or spending most of its time tending to one side or against one end stop

The Logger feature works with all Seakeeper models except M-Series and is best for determining the causes of erratic precession. Seakeepers with ConnectBox and GCM software 11.130 or later create data logs manually or after alarms, which can be retrieved per SWI-137—ConnectBox Data Log Retrieval.

It is good engineering practice to first take a short (15-second) test set of logs and save them to a desired location. This test run will verify that the logs are saved without any software issues before starting an extended test run.

1. To capture CAN messaging in logger function, **SELECT** UTILITIES from HOME screen.
2. **SELECT** LOGGER button on UTILITIES screen.
3. **SELECT** appropriate data for each dropdown.
4. **ENTER** notes to document how and when Seakeeper is not performing as expected, case number, and vessel name.
5. Once all fields populated, **SELECT** NEXT button.
6. **SELECT** save location for LOGGER file to be recorded.

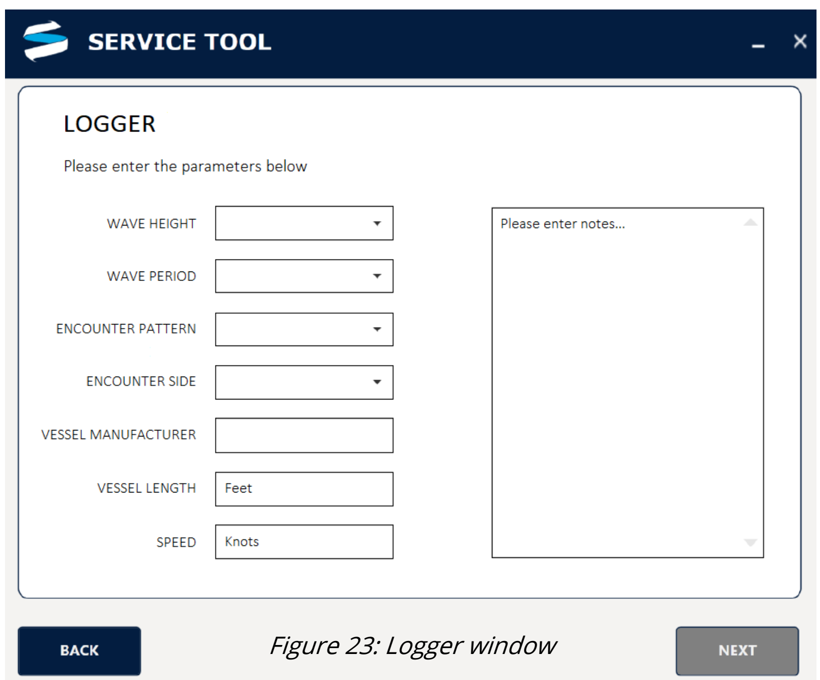


Figure 23: Logger window

7. **PERFORM** a sample run as follows:

    a. **PRESS** RECORD.

    b. After 5 seconds, **PRESS** STOP.

    c. **NAVIGATE** to location where file saved.

    d. **VERIFY** file created with "CANLOG" and a time stamp.

8. <u>**IF**</u> sample file not created,
    <u>**THEN**</u>:

    a. **SELECT** BACK button.

    b. **REENTER** all data for vessel and sea conditions.

    c. **SELECT** NEXT button.

    d. **RETRY** sample run per step 7 above.

9. **SELECT** UPDATE CAN FRAMES to allow scrolling data (Figure 24).

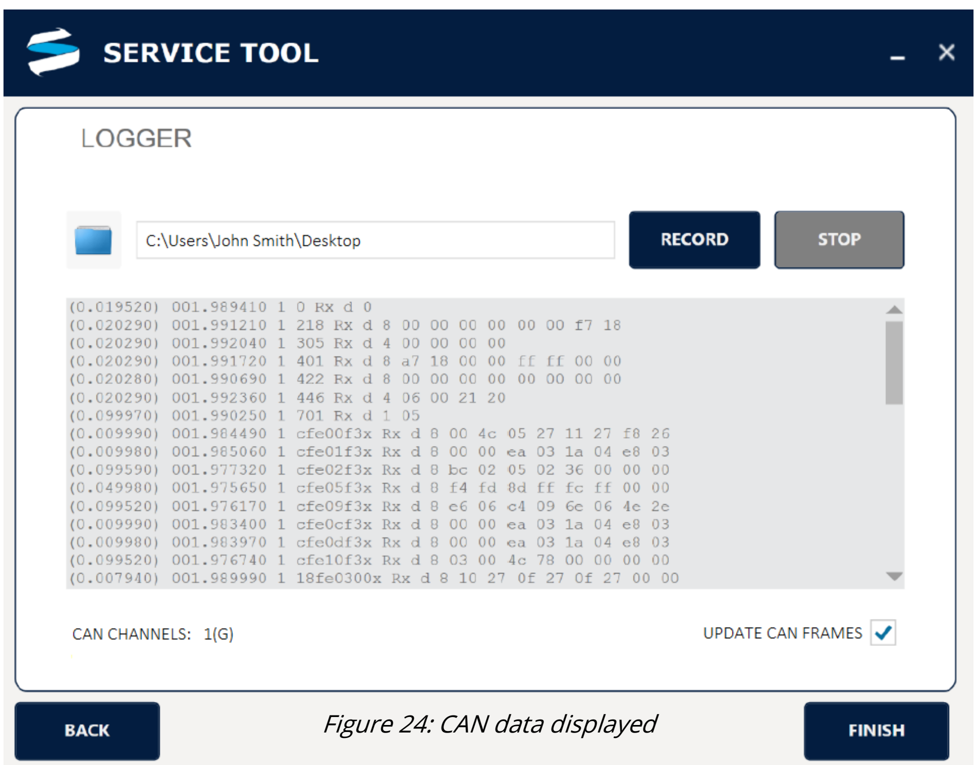


Figure 24: CAN data displayed

10. **SELECT** RECORD button to log CAN traffic.

11. **NOTE** time of start.

12. **ENSURE** numbers begin scrolling in window.

NOTE:
If PC enters SLEEP mode while recording data, the data will be lost.

13. **WHILE** recording data file,
    ENSURE PC does NOT enter SLEEP mode.

14. **IF** Seakeeper operation becomes erratic,
    THEN **NOTE** time of occurrence in minutes after starting recording.

15. Once data recorded, **SELECT** STOP button to stop recording.

    a. **IF** collecting data to evaluate poor roll reduction performance,
       THEN **TAKE** two sets of logs: one with Seakeeper UNLOCKED and second while LOCKED.

16. **EMAIL** saved log files and times of improper operation to support@seakeeper.com for review.

6.2 PVMO

The PVMO (Proportional Valve Manufacturer's Offset) is a value of current applied to the brake valve during precession by the GCM. On specific Seakeeper models, if the proportional valve must be replaced, a new PVMO number will be provided with the replacement valve. If only the GCM were replaced, the previous PVMO value would be needed. PVMO functionality may be restricted based on the license access level granted.

1. From UTILITIES screen, **SELECT** PVMO button.

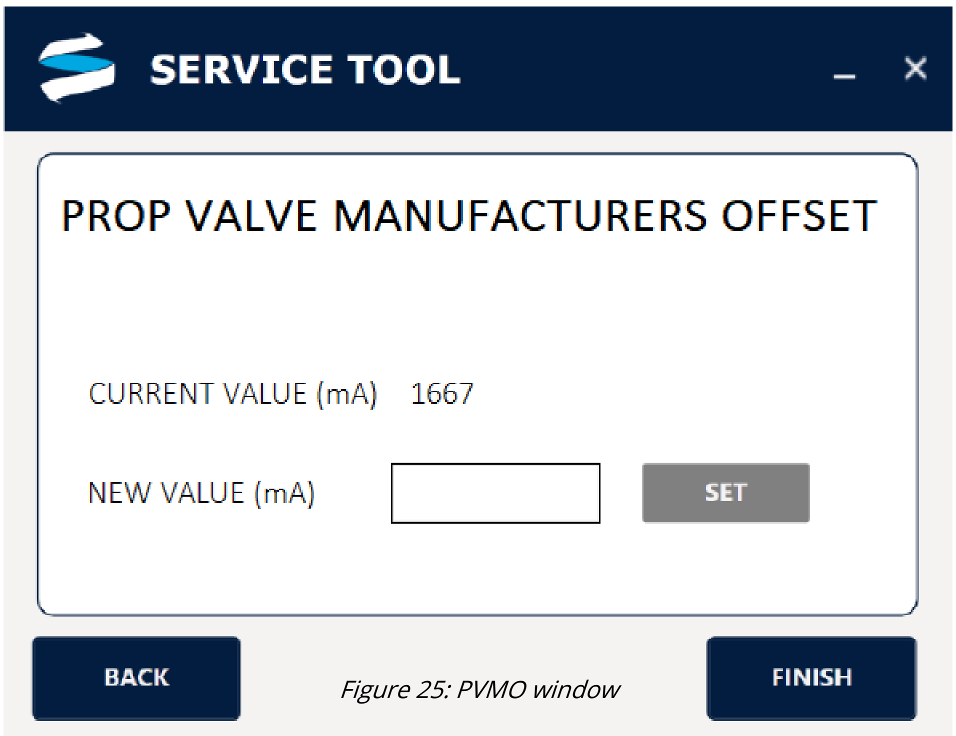


Figure 25: PVMO window

2. In NEW VALUE field (Figure 25), **ENTER** Seakeeper-provided PVMO value.

3. **PRESS** SET button.

4. **CONFIRM** CURRENT VALUE field updated to new value.

6.3 PROGRAM

This utility option requires a higher access level, which can be granted when needed. Those given the program functionality will receive custom files to program the GCM. The file will have a ".sreb" file name ending.

1. From UTILITIES screen, **SELECT** PROGRAM button.

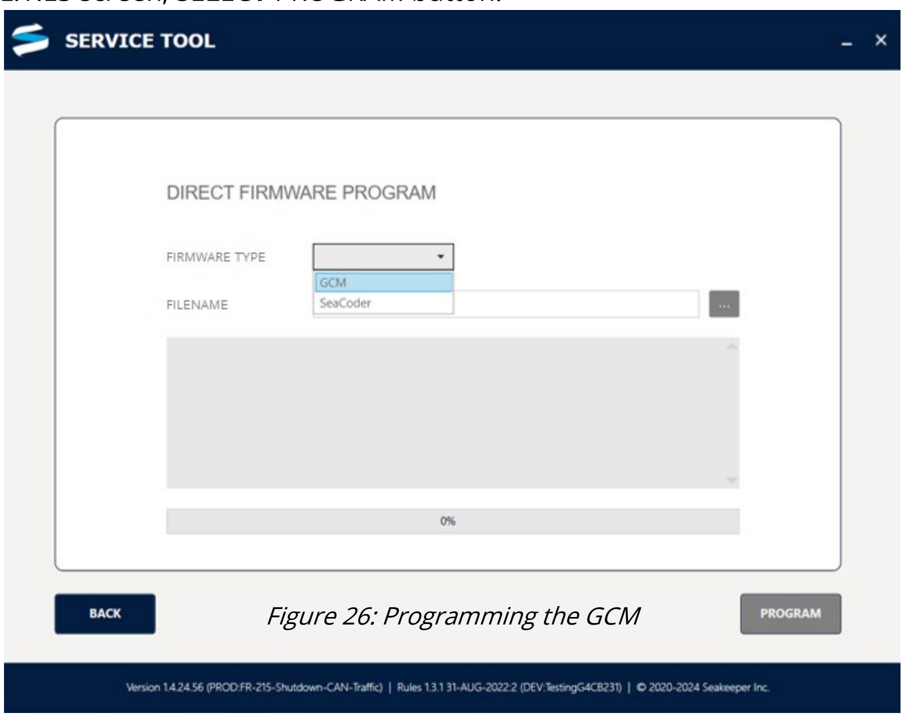


Figure 26: Programming the GCM

2. **SELECT** Firmware Type: GCM from dropdown menu.
3. **SELECT** the ellipsis (three dots) to right of filename window to select the GCM file to program. [**NOTE:** Filename will end in ".sreb."]
4. **SELECT** appropriate file.
5. **SELECT** PROGRAM button.
6. Once GCM programmed, **SELECT** FINISH.

*********** END ***********

REVISION	DESCRIPTION	APPROVAL	DATE
5	Added features of GCM and ConnectBox software release of August 2024.	A Patricio	08OCT2024
6	Add SWI-137, ConnectBox Data Log Retrieval reference for data logs created by ConnectBox. Minor editorial corrections.	A Patricio	19MAR2025